

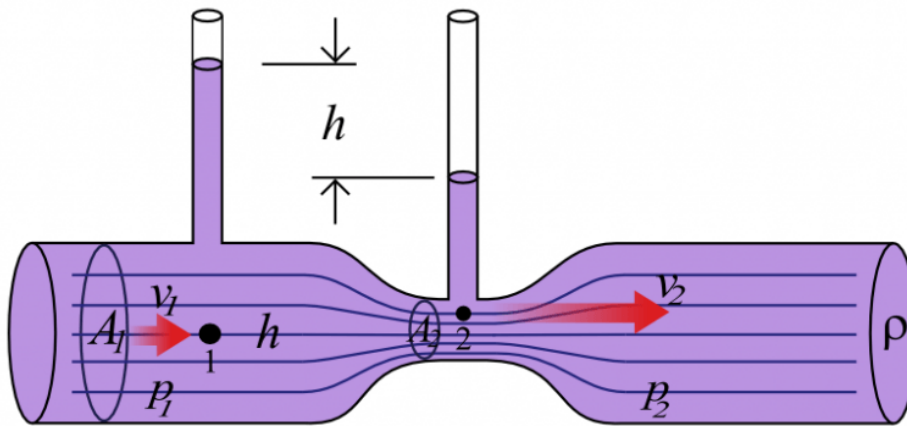


**Mechanical Engineering Department
of Cankaya University**



ME414 Mechanical Engineering Laboratory II

Venturi Meter Experiment



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VENTURI METER EXPERIMENT

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ABSTRACT: In this experiment, firstly, studies were conducted in which we could learn what the venturi meter is and what it does. Later, patent studies and detailed literature researches were conducted on this subject. The main parts of the venturi meter were examined. As a result of this examination, it was determined that it consists of 3 main parts. The first of these main parts is the Converging Part, the second is the throat and the third is the Diverging Part. Then, how the venturi meter works and what areas it is used in were investigated. After these priority researches were completed, the calculation process was started. Calculate the calculations using Bernoulli's equation ($\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2$) and the theoretical flow rate using the Continuity equation ($Q = V_1 A_1 = V_2 A_2$) value was calculated. Then, using the actual flow rate value given to us, the discharge coefficient value was calculated. After performing these calculations, a venturi meter was designed in the Autodesk Inventor drawing program with the data given to us and the values we found. After completing the venturimeter design, detailed flow analysis was performed using the Ansys program. We made our flow analysis for different geometries and diameters. In this way, we were able to compare them. We clarified the accuracy of our results.

Keywords: Venturi Meter, Experiment, Bernoulli, Continuity, Autodesk Inventor, Ansys

I. INTRODUCTION

Venturi meter is a measuring instrument used to measure the flow rate of fluids, open at both ends, narrowing at both ends towards the middle of the cross section and in the form of a thin tube. While the fluid velocity reaches the highest value in the narrowest section of the tube forming a throat by tapering, the fluid pressure decreases, and a pressure difference occurs between the wide section and the narrow section of the tube. This pressure difference is measured with the help of the "Closed manometer" connected to the inlet and outlet of the venturi meter. With the help of the pressure values obtained because of this measurement, the flow rate of the fluid passing through the tube is determined. "Bernoulli equation" and "Continuity Equation" are used to calculate the mass or volumetric flow rate.

Venturi meter is a laboratory instrument and is used in studies related to fluids. According to the research, Özcan Tuncay examined the 'Parameters affecting the efficiency of liquid injection in venturi meters' [1].

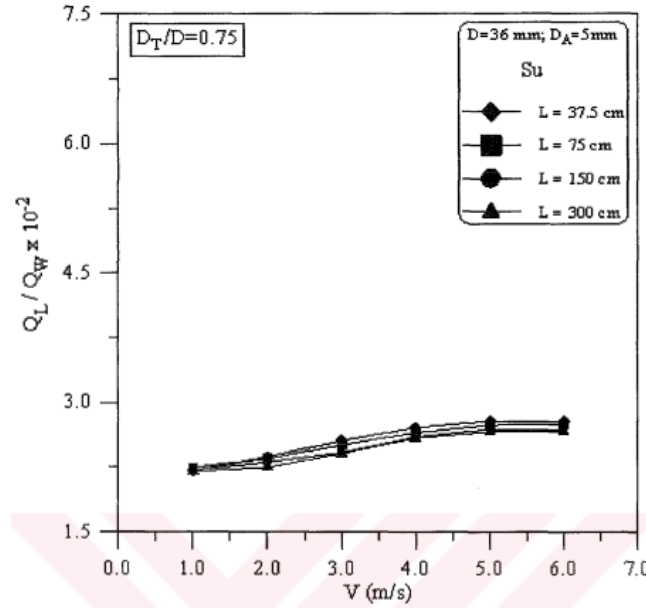


Figure 1: The Graph of The Variation of The Injection Performance in Ventures of Different Diameters with 2.5 mm and 5.00 mm Suction Pipes with a Constriction Ratio of 0.75

In this study, the efficiency of the parameters affecting the injection performance in injection systems using venturi was investigated. For this purpose, venturis with three different inlet diameters and two diameters in the throat area of venturi were used. The efficiency of diameter ratios for each inlet diameter was also examined. In addition, the effect of the suction pipe diameter, the velocity of the fluid at the venturi inlet, the length of the tube after the venturi outlet and the viscosity of the fluid used for injection on the injection performance are among the parameters investigated. Gürçan Gürsoy, in his study named 'Experimental investigation of pressure loss in Venturi meter and nozzle'[2] due to the geometries of venturi meters, the head losses are less than other flow meters. Especially in the testing of propulsion systems in airplanes, in systems with liquid gas mixture as a fluid, venturi is widely used in the measurement of flow in the plastic paint, paper and leather industry.

Control and Embedded Systems Research Group students researched the project named 'Air – water two-phase flow measurement using a Venturi meter and an electrical resistance tomography sensor' in China.[3] The aim of this research is to develop a new flowrate measurement method for air–water two-phase flow by combining a Venturi meter and an ERT sensor. In order to minimize the influence of flow pattern on the measurement, the flow pattern is identified in real time using the ERT sensor incorporating the Least Squares Support Vector Machine (LS-SVM) classification method. In order to acquire the mass quality, the void fraction is firstly measured using the ERT sensor and the LS-SVM regression method, and then the quality is calculated from the void fraction through the void fraction-quality correlation. The mass flowrate of the air–water flow is calculated through the flowrate measurement correlation.

II. EXPERIMENTAL TEST SET-UP & DATA PROCESSING

II.I EXPERIMENTAL TEST SET-UP

In this experiment, we made 2 flow analyzes from Ansys program according to different measurements. The difference in these analyzes is due to the geometry. In our first analysis, the venturimeter is a closed system, while in the second analysis, our system is open to the atmosphere in critical regions.

First, we made a 3D venturimeter drawing from the Autodesk Inventor program according to the measurements given in this experiment. After first determining the circles of the venturimeter, I created the main part of the venturimeter by applying the loft procedure in this program. We made the extrude process by drilling the pressure head length on the venturimeter according to the given values and taking the H1 value as 305 mm and the H2 value as 125 mm. In this way, we created a setup that will cover all H values.

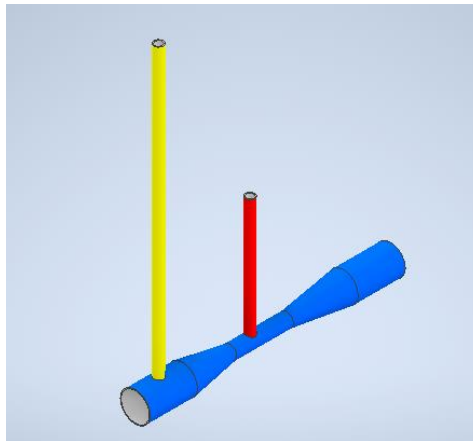


Figure 2: Venturimeter in Autodesk Inventor

Secondly, we drew our 2D geometry in the Ansys program. We have defined 25 mm inlet diameters, 10 mm throat and 35 mm outlet diameters in our geometry. We made sure that the lines drawn in geometry are parallel at the entrance and exit. Thanks to the lines thrown into geometry, we divided our geometry into 4 parts. In this way, we obtained 4 different zones. These lines are imaginary lines. It was used to separate the geometry to 4 parts. It does not belong to the part.

After completing our operations in the geometry area, we switched to the mesh area. After entering the mesh, we selected the CFD section from the physics preference area. Then we created our first mesh draft with the generate mesh command. Since this mesh is not enough for an accurate analysis, we have provided a better mesh with face meshing. By dimensioning our geometry (from the sizing tab), we put an equal number of points on our lines in each region. In this way, we were able to create a tighter mesh area in the small area. At the end of these processes, we have observed the tightness of the excess mesh.

Finally, we determined the input and output parts with the name selection command and completed the mesh part. After the mesh part, we switched to the setup part. If the Mach number is less than 1, that is, if our flow is incompressible flow, we should have made a pressure-based solution.

In our system, we used density based because the number of Mach is greater than 1. Since our solution is not time dependent, we chose the steady state solution. Turning the Gravity on or not would have no effect on the problem. Turning on the gravity would increase my computation cost. We did not turn on gravity because we wanted to reduce our computation cost. If there is a heat change, that is, if there is heat transfer, we need to open the energy equation. Since changes in temperature and pressure will occur, we opened the energy equation. We turned the density into ideal gas. Because the temperature change would affect the density at the temperature that would be, we had to turn the viscosity in the same way. Because our system was in a position to be affected by viscous effects. That is why we chose Sutherland's viscosity, so the density and viscosity varied with temperature change. The change in pressure and velocity affected the temperature according to the formula ' $Pv = nRT$ '. We also observed a change in density as density is a function of temperature. In Boundary conditions, we entered the pressure inlet and pressure outlet issue, and then we defined the entrance pressure as 5 bar. We defined our pressure in the outlet as gage pressure since it is open to the atmosphere. We chose the inlet as the reference value. Since the pressure value we entered the system was an ideal gas equation, the program could calculate its speed. In this way, we were able to calculate coefficient directly from the wall section. Then we came to the initialization section and selected the standard initialization box in this section. We chose the inlet as the reference section and initialized the program. Finally, we completed our analysis from the iteration section of the run calculation section. We obtained the graph as a descending graph, so our values converge. Then we looked at the variable speed. There is a boundary layer at all wall edges. The boundary layer is the area where viscous effects are important. Our speed starts at zero and reaches u-infinity.

The speed at the entrance increased over time. It increased a little more in the middle, made a very rapid debut on the exit. The decrease in pressure caused an increase in speed. Finally, we chose our contour variable temperature. We observed that our temperature values decreased at the exit. We have seen that the system also provides temperature control.

In addition to all these analysis and design, we produced the Venturi Meter in a simple way. And we got a test with the water and the system worked well. Thus, we observed the operation of the system.



Figure 3: Venturi Meter Experiment Setup

Task ID	Task Name	Start Date	End Date	Duration (In Days)	18/03/2021	25/03/2021	01/04/2021	01/04/2021	08/04/2021	08/04/2021	15/04/2021	15/04/2021	22/04/2021	22/04/2021	29/04/2021
T1	Literature, Mathematical Equations, Morphological Chart, Gantt Chart	18/03/2021	01/04/2021	14											
T2	Determining Data, Concept Design and Solid Model Design	01/04/2021	08/04/2021	7											
T3	Calculation Operations	08/04/2021	15/04/2021	7											
T4	Fluid Analysis and Calculations	15/04/2021	22/04/2021	7											
T5	Creating Graphics and Conclusion	22/04/2021	29/04/2021	7											

Figure 4: Gantt Chart

This is our Gantt chart. We will perform this experiment for 6 weeks. It is shown how long the studies were done in this chart.

Morphological Chart	Solutions	
Direction of the Fluid	In the Direction of Gravity ↓	Perpendicular to Gravity ↓
Cross Section Area Inlet and Outlet	Large ↓	Narrow ↓
Liquid Type	Water ↓	Oil ↓
Flow Type	Laminar ↓	Turbulent ↓
Cross Section Area in Throat	Large ↓	Narrow ↓
Throat Angle	Sharp Transitive ↓	Soft Transition ↓
Pressure Difference Measurement Method	Manometer Connected ↓	Toricelli Method ↓
Environmental Pressure	Open Air Pressure in High Places ↓	Open Air Pressure at Sea Level ↓

Figure 5: Morphological Chart

This is our morphological chart. We have two concepts here. One of them is the blue concept. The other is green concept. For example, while there is a manometer in the blue concept, there is open-air pressure in the green concept. Our best concept is the green concept. The analyzes, designs and calculations we have carried out have been made according to this concept.

II.II DATA PROCESSING

In this experiment, certain data were given to us without any calculations. Dividing the given volume values by the given time values, 9 units of flow were found. This value is our actual flow rate.

At the same time, theoretical flow rate values were calculated from the 9 head values given. In these calculations, the fields are kept constant. Then, the discharge coefficient was calculated by the ratio of the calculated flow rate values to each other.

H1 (mm)	H2 (mm)	Time (sec)	Volume (L)	Flow Rate (m ³ /s)	Flow Rate (L/m)
125	119	25.82	1	$3,872 \times 10^{-5}$	2,3237
140	125	22.89	2	$8,737 \times 10^{-5}$	5,2424
145	120	26.92	3	$11,144 \times 10^{-5}$	6,6864
170	105	18.72	3	$16,025 \times 10^{-5}$	9,6153
195	110	14.09	3	$21,291 \times 10^{-5}$	12,77
220	100	17.73	5	$28,2007 \times 10^{-5}$	16,92
240	95	34.66	10	$28,851 \times 10^{-5}$	17,311
280	75	29.97	10	$33,3667 \times 10^{-5}$	20,02
305	70	26.35	10	$37,993 \times 10^{-5}$	22,77

Table 1: Table of Trial of Experimental Data

III. NUMERICAL METHODOLOGY

Using the data given, we calculated the theoretical flow rate values from Bernoulli Equation. While calculating the theoretical flow rate values, we kept the areas constant. As a result of the calculations made, we observed that as the pressure head difference increased, the flow rate value also increased.

Assume incompressible flow and no frictional losses, from Bernoulli's Equation.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2 \quad (1)$$

Use of the continuity equation,

$$Q = V_1 A_1 = V_2 A_2 \quad (2)$$

equation (1) becomes.

$$\frac{P_1 - P_2}{\rho g} + Z_1 - Z_2 = \frac{V_2^2}{2g} \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right] \quad (3)$$

$$Q_{theoretical} = A_2 V_2 = \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}} \sqrt{2g \left(\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right)} \quad (4)$$

$\left(\frac{P_1 - P_2}{\rho g} + (Z_1 - Z_2) \right)$ term represents the difference in piezometric head (Δh) between the two sections 1 and 2. The above expression for V_2 is obtained based on the assumption of one-dimensional frictionless flow. Hence the theoretical flow can be expressed as:

$$Q_{theoretical} = A_2 V_2 = \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}} \sqrt{2g(\Delta H)} \quad (5)$$

Thus;

$$Q_{theoretical} = \frac{\sqrt{2g(\Delta H)}}{\sqrt{\left(\frac{1}{A_2^2} - \frac{1}{A_1^2} \right)}} \quad (6)$$

Because of the above assumptions, the actual flow rate, Q_{act} differs from $Q_{theoretical}$ and the ratio between them is called the discharge coefficient, C_d which can be written as

$$C_d = \frac{Q_{act}}{Q_{the}} \quad (7)$$

The discharge coefficient is always less than due to various losses (friction losses, area contraction etc.). Where C_d is an adjustment factor called the discharge coefficient, which you can find by experiment. It is usually between 0.92 to 0.99.

Trial1:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(125 - 119) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 6,23 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial2:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(140 - 125) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 9,9 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial3:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(145 - 120) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 12,71 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial4:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(170 - 105) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 20,50 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial5:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(195 - 110) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 23,5 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial6:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(220 - 100) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 27,9 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial7:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(240 - 95) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 30,6 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial8:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81(280 - 75) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2}\right)}} = 36,41 \times 10^{-5} \text{ m}^3/\text{s}$$

Trial9:

$$Q_{\text{theoretical}} = \sqrt{\frac{2 \times 9,81 (305 - 70) \times 10^{-3}}{\left(\frac{1}{(1,77 \times 10^{-4})^2} - \frac{1}{(7,92 \times 10^{-4})^2} \right)}} = 38,9 \times 10^{-5} \text{ m}^3/\text{s}$$

H1 (mm)	H2 (mm)	Sqaure root of difference in pressure head	Q Actual $10^{-5} \times \text{m}^3/\text{s}$	Q Theoretical $10^{-5} \times \text{m}^3/\text{s}$	Cd
125	119	2,44	3,872	6,23	0,6215
140	125	3,87	8,737	9,9	0,8825
145	120	5	11,144	12,71	0,8767
170	105	8,62	16,025	20,5	0,7817
195	110	9,21	21,291	23,5	0,9060
220	100	10,95	28,2007	27,9	1,0107
240	95	12,04	28,851	30,6	0,9428
280	75	14,31	33,3667	36,4	0,9166
305	70	15,32	37,993	38,9	0,9766

Table 2: Table of Calculated Trial of Experimental Data

Comment: The value closest to 1 among the discharge coefficient values we have found from the ratio of real and experimental flow rate values is the most accurate solution of our experiment. This value is shown in bold in the table.

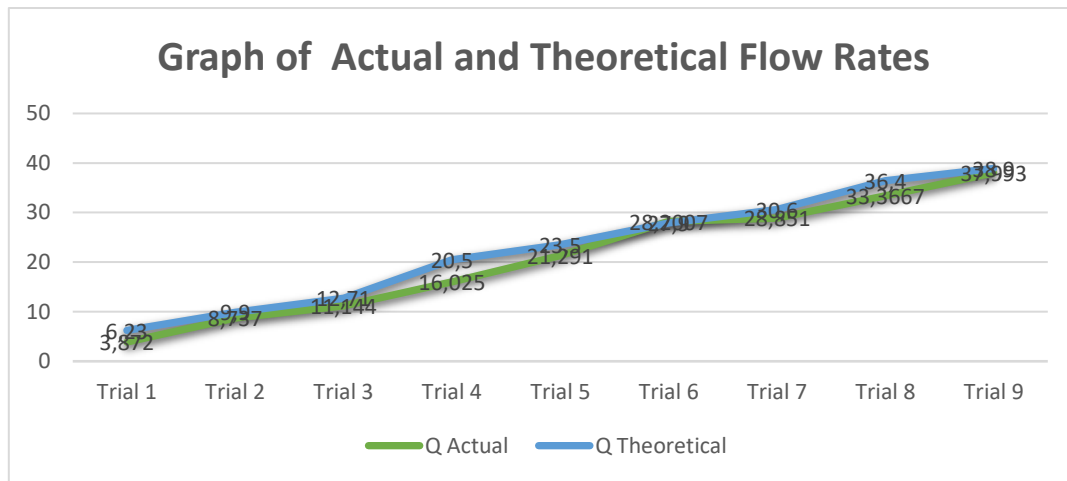


Figure 6: Graph of Actual and Theoretical Flow Rates

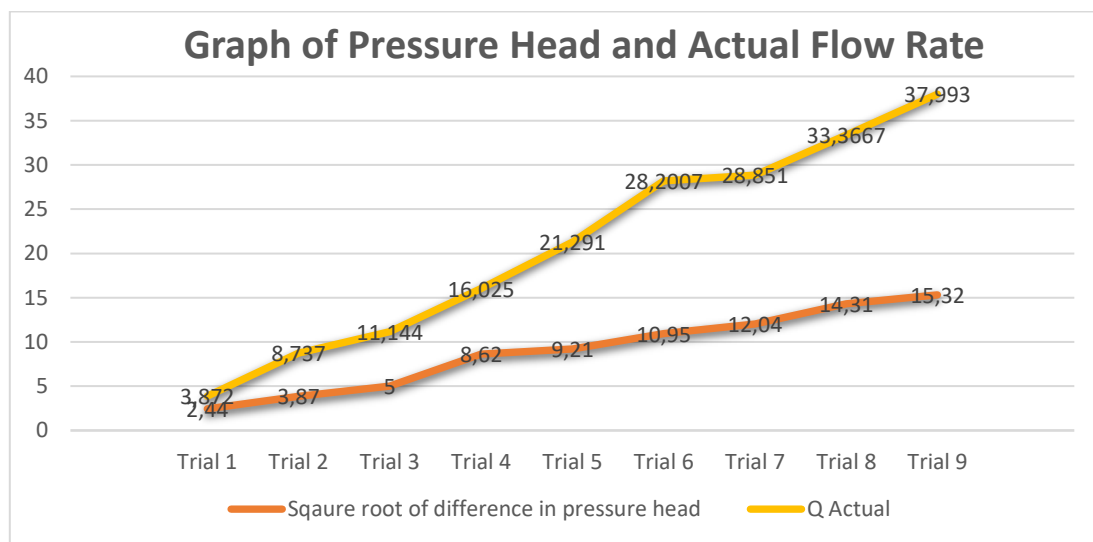


Figure 7: Graph of Pressure Head and Actual Flow Rate

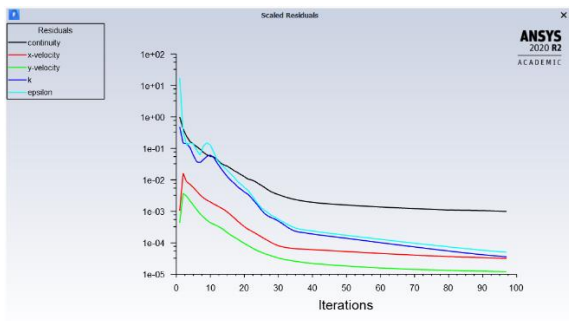
The data received were recorded one by one. In the flow rate equation, we deduced from the Bernoulli equation, the flow rate was calculated separately for each data. The calculated flow rates are the results we found theoretically. Actual flow rates are given in the table (-). We found the discharge coefficient by comparing the actual flow rate data and the theoretical flow rate data. The value closest to 1 of these discharge coefficient values we found is the best measured and calculated flow rate value for us. This value was reached in the **6th** trial.

The reason why other trials are far from 1 may be due to the error of the person making the measurement, the error of the person making the calculation, or the lack of correlation of the measuring instruments.

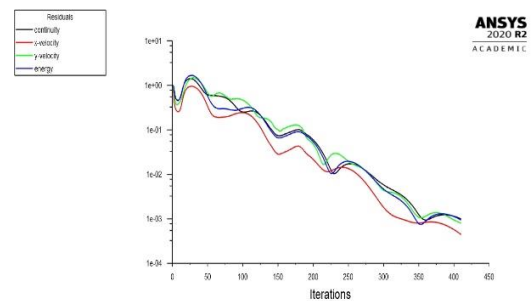
IV. RESULTS AND DISCUSSION

We wanted to observe how our results change with geometry by choosing different geometries for experimental analysis. In our first comparison, the differences created by two venturi meters with completely different geometries are observed. In other comparisons, our geometries are the same, but our entrance and exit cross-sectional areas are equal, large, and small, and changes are analyzed according to the parameters we have determined. The results of these analyzes were compared and interpreted.

Comparison of Venturi Meters with Different Geometries and The Same Diameter Inlet and Outlet:



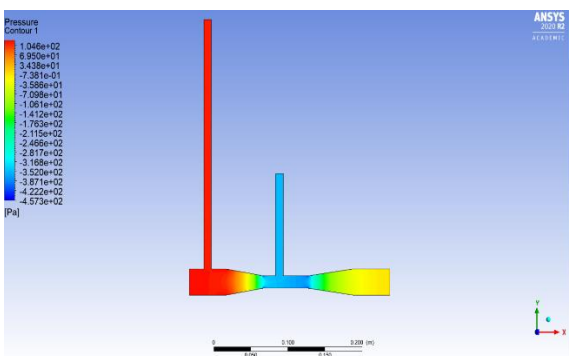
A-) Venturi Meter with open air pressure



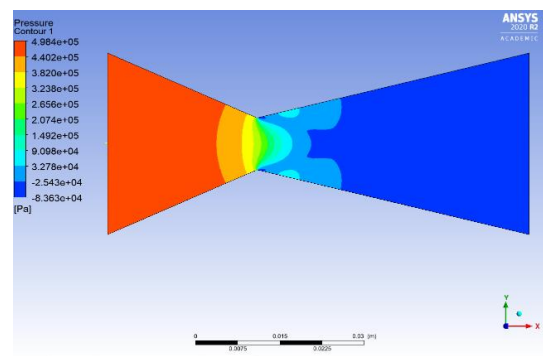
B-) Venturi Meter without Pressure Head

Figure 8: Residual Graphs Different Geometries with Equal Inlet and Outlet Diameters

Comment: Both of these residual plots were supposed to converge in the analysis. Our analysis also converts these charts.



A-) Venturi Meter with open air pressure

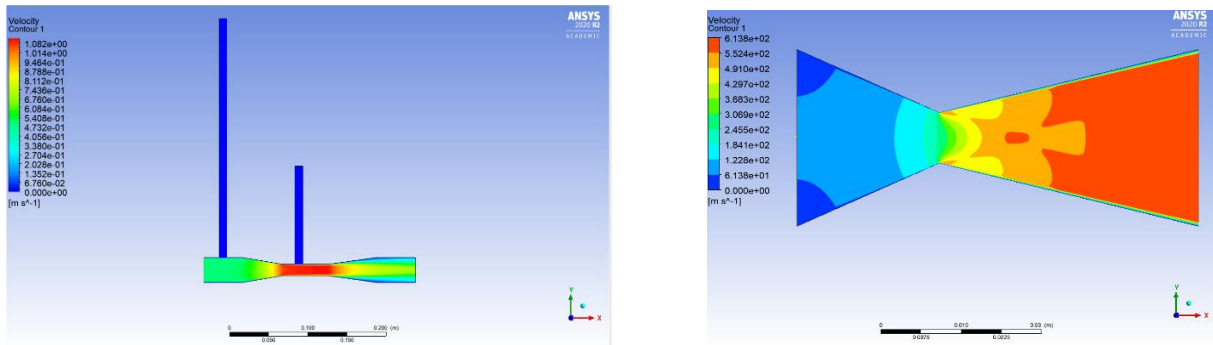


B-) Venturi Meter without Pressure Head

Figure 9: Pressure Contours Different Geometries with Equal Inlet and Outlet Diameters

Comment: In the flow analysis, our pressure at the inlet in the pressure contour part is high for both geometries. The pressure in the throat part decreases a little. However, while the pressure increases in the venturi meter, which has open air pressure at the outlet, the pressure decreases even more in the venturi meter without pressure heater.

Thanks to the Throat, while the pressure difference is low in the open-air pressure venturimeter, the pressure difference is observed too much in the venturimeter, which does not have a pressure heater, because there is no throat.

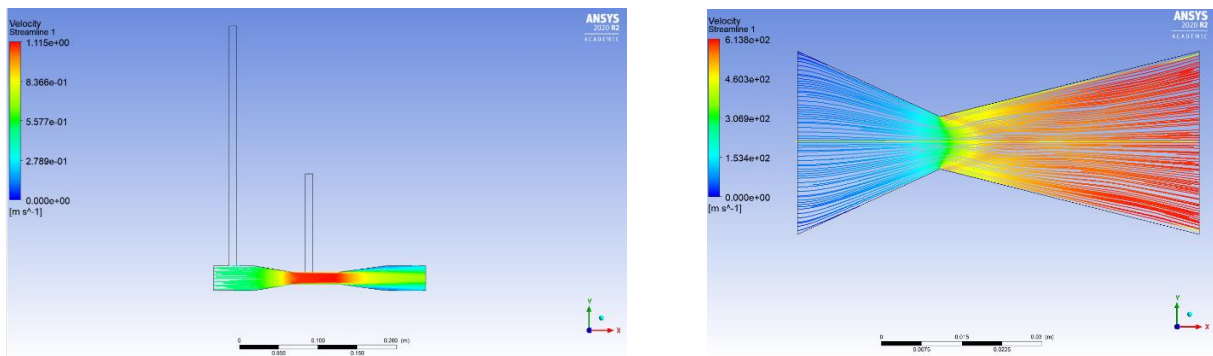


A-) Venturi Meter with open air pressure

B-) Venturi Meter without Pressure Head

Figure 10: Velocity Contours Different Geometries with Equal Inlet and Outlet Diameters

Comment: Speed is inversely proportional to pressure. This is because in the formula $Pv = nRT$, when P increases, v decreases because the opposite side is constant. In venturi meter with open air pressure when the speed is normal at the entrance. Speed has increased considerably in throat. Since this venturi meter has a throat, there is no loss of speed at the throat line. In the exit part, our speed has decreased. On the other hand, in venturi meter without pressure sensor, it increased in continuation due to the narrowing cross-section when the velocity was low at the entrance.

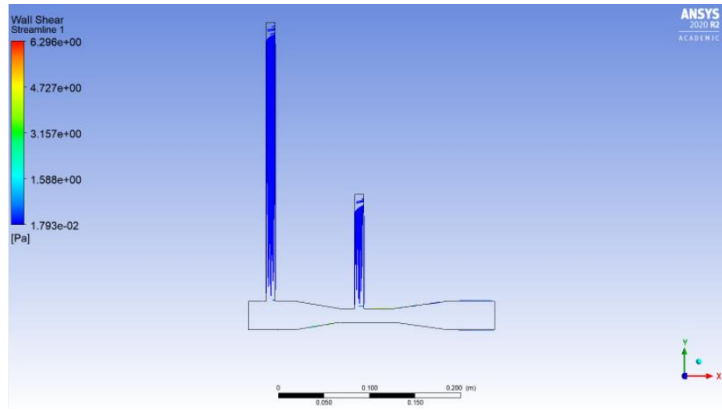


A-) Venturi Meter with open air pressure

B-) Venturi Meter without Pressure Head

Figure 11: Velocity Streamlines Different Geometries with Equal Inlet and Outlet Diameters

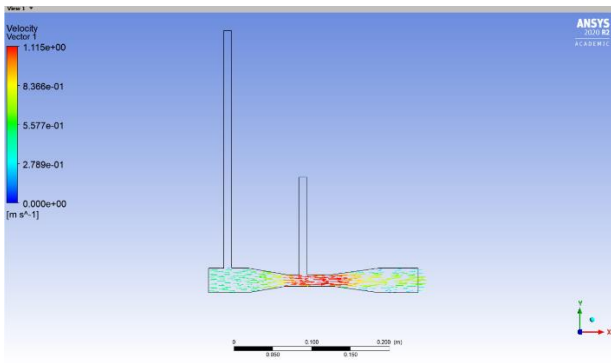
Comment: Since the speed of the system is sufficient in Velocity Streamlines, no vortex was observed.



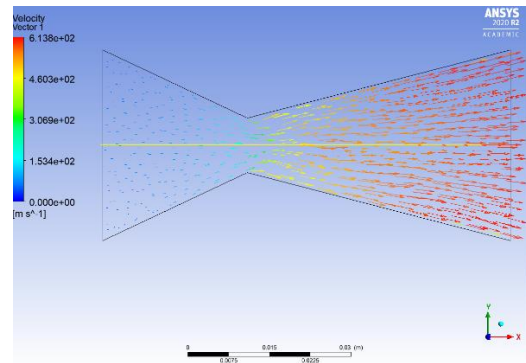
A-) Venturi Meter with open air pressure

Figure 12: Wall Shear Streamline Equal Inlet and Outlet Diameters

Comment: Pressure head values are observed in the figure.

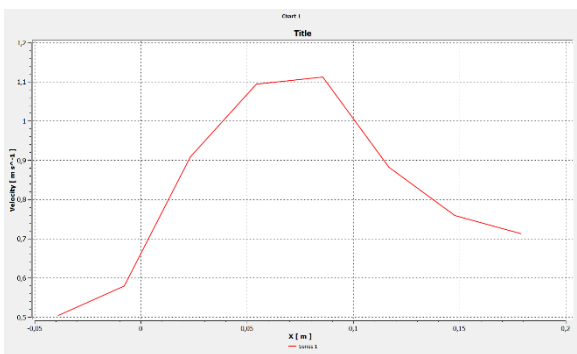


A-) Venturi Meter with open air pressure

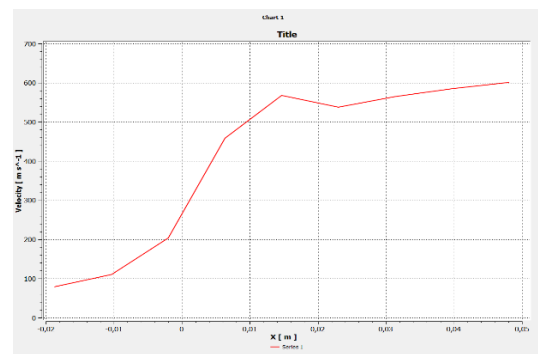


B-) Venturi Meter without Pressure Head

Figure 13: Velocity Vectors Different Geometries with Equal Inlet and Outlet Diameters



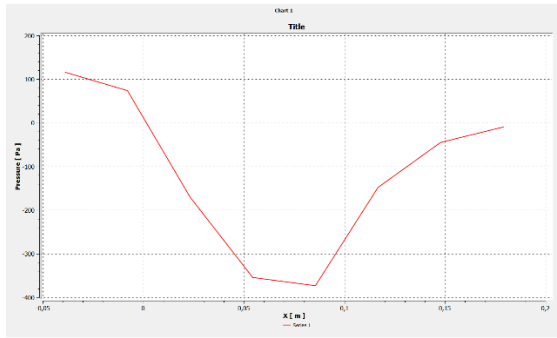
A-) Venturi Meter with open air pressure



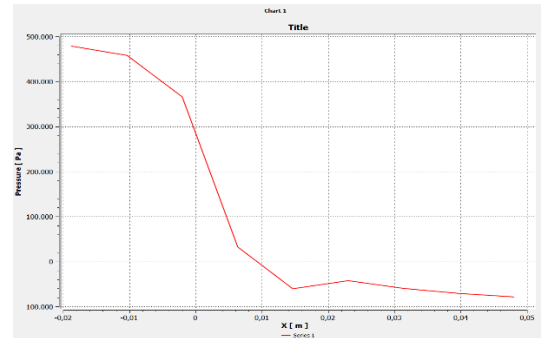
B-) Venturi Meter without Pressure Head

Figure 14: Velocity Charts Different Geometries with Equal Inlet and Outlet Diameters

Comment: Since the open-air pressurized venturi meter has a throat, the speed increased up to the throat and remained constant throughout the throat. After the throat, the speed decreased. Since the venturimeter without pressure head does not have a throat, the velocity continued to increase towards the outlet due to narrowing cross-section.



A-) Venturi Meter with open air pressure

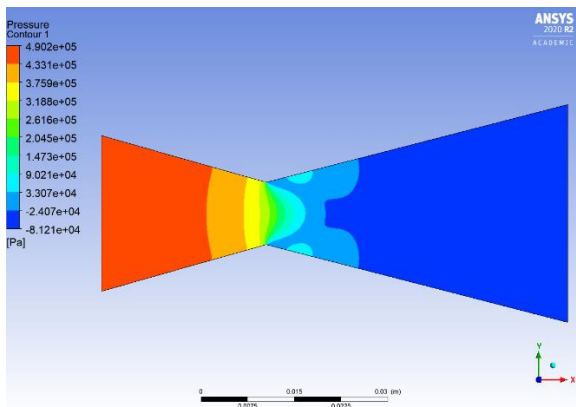


B-) Venturi Meter without Pressure Head

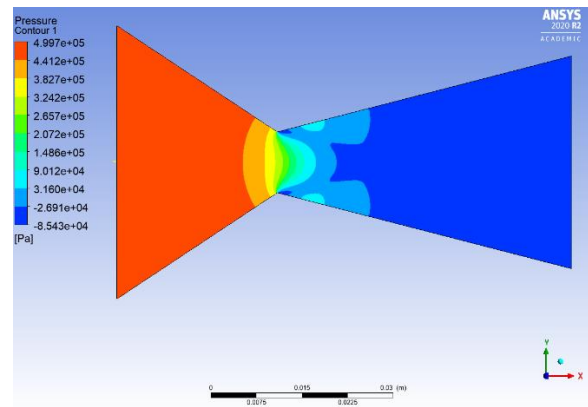
Figure 15: Pressure Charts Different Geometries with Equal Inlet and Outlet Diameters

Comment: Pressure and velocity are inversely proportional to each other due to the ideal gas equation. Therefore, the graphs are exactly the opposite of each other.

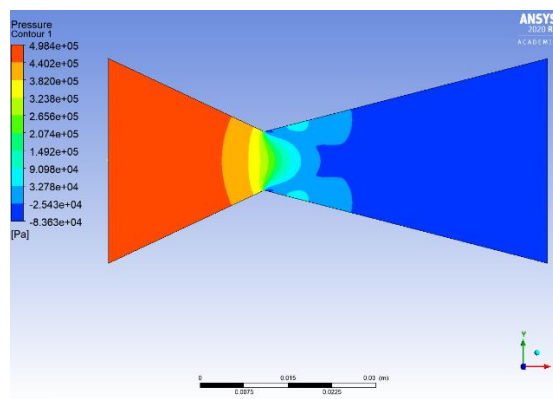
Comparison of Venturi Meters with Different Inlet and Outlet Diameters:



A-) Small Inlet Diameter



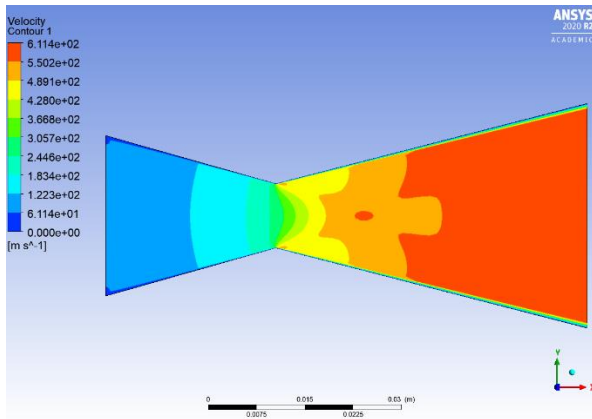
B-) Large Inlet Diameter Small Outlet Diameter



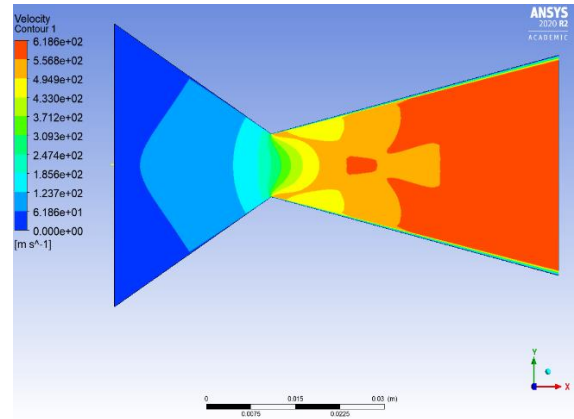
C-) Equal Inlet and Outlet Diameter

Figure 16: Pressure Contours Different Inlet and Outlet Diameters

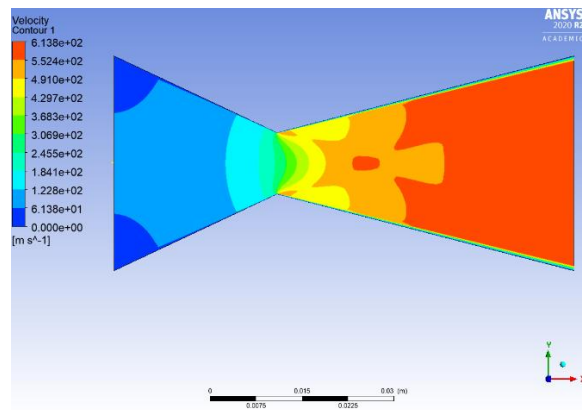
Comment: When the constant pressure at the entrances encountered a wider cross section, the pressure change started to occur earlier.



A-) Small Inlet Diameter



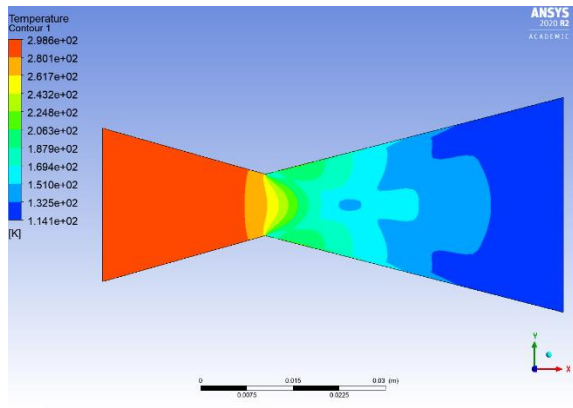
B-) Large Inlet Diameter Small Outlet Diameter



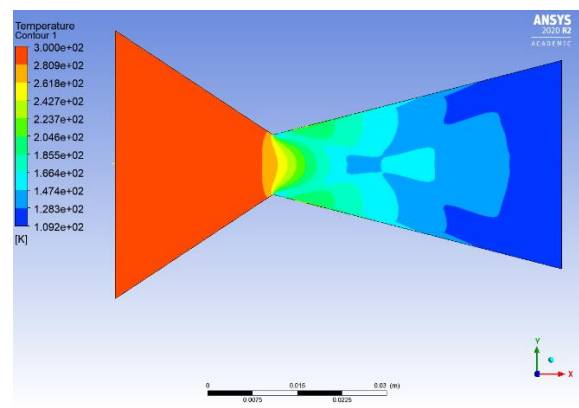
C-) Equal Inlet and Outlet Diameter

Figure 17: Velocity Contours Different Inlet and Outlet Diameters

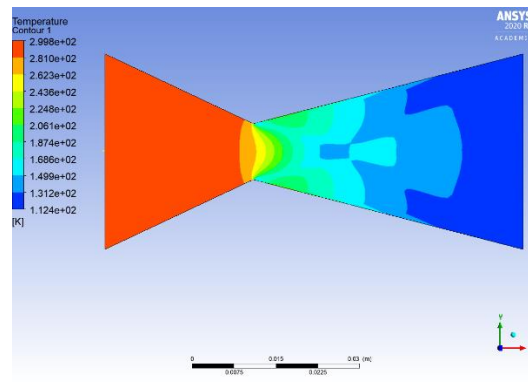
Comment: Due to the constant pressure at the entrances, according to the continuity equation ($A_1 V_1 = A_2 V_2$), the inlet velocity has changed inversely with the cross-sectional area.



A-) Small Inlet Diameter



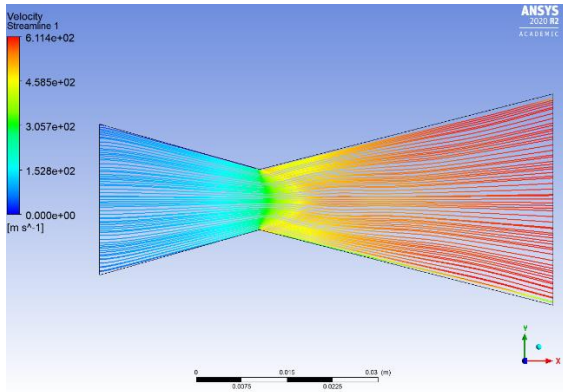
B-) Large Inlet Diameter Small Outlet Diameter



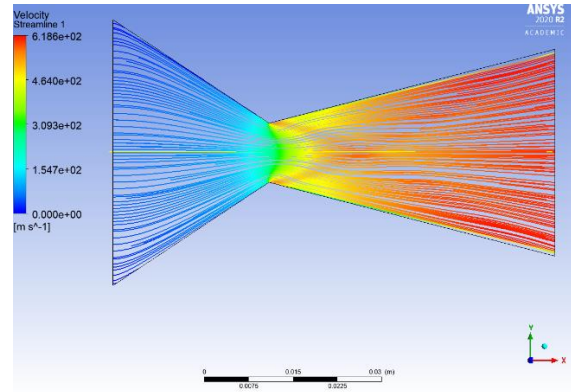
C-) Equal Inlet and Outlet Diameter

Figure 18: Temperature Contours Different Inlet and Outlet Diameters

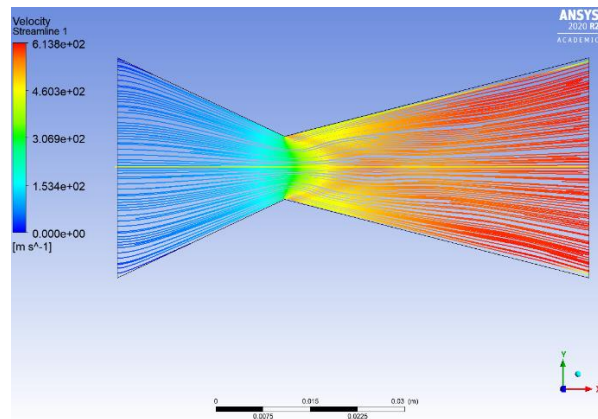
Comment: According to the ideal gas equation [$PV = nRT$], the temperature has changed depending on the pressure and speed change.



A-) Small Inlet Diameter



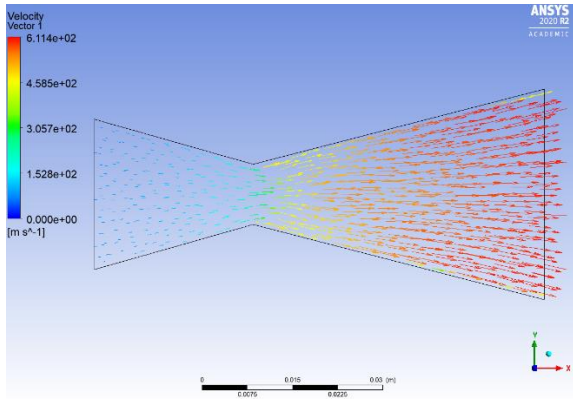
B-) Large Inlet Diameter Small Outlet Diameter



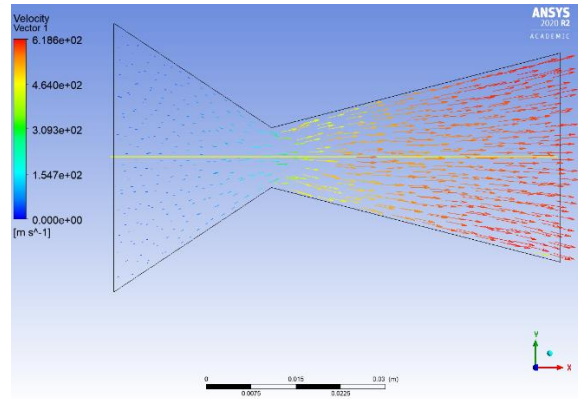
C-) Equal Inlet and Outlet Diameter

Figure 19: Velocity Streamlines Different Inlet and Outlet Diameters

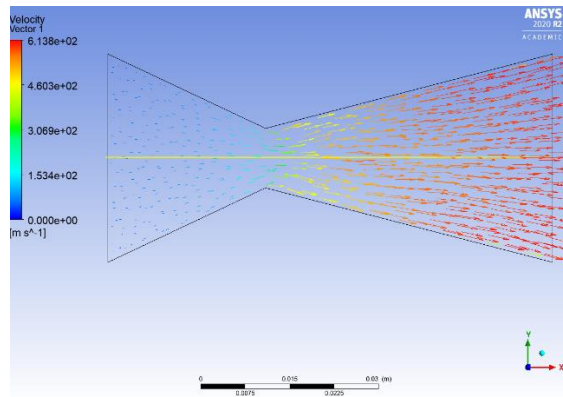
Comment: No vortex formation was observed in the system as there was sufficient speed.



A-) Small Inlet Diameter

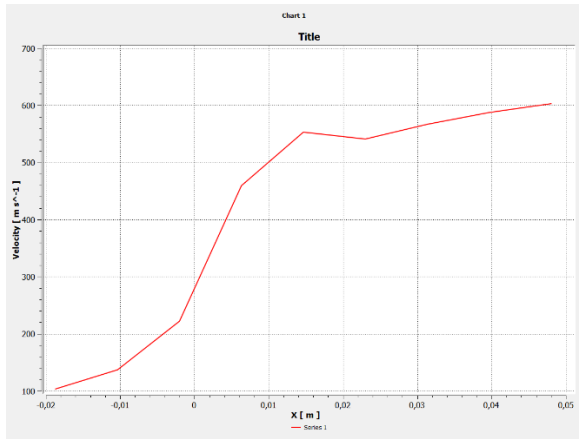


B-) Large Inlet Diameter Small Outlet Diameter

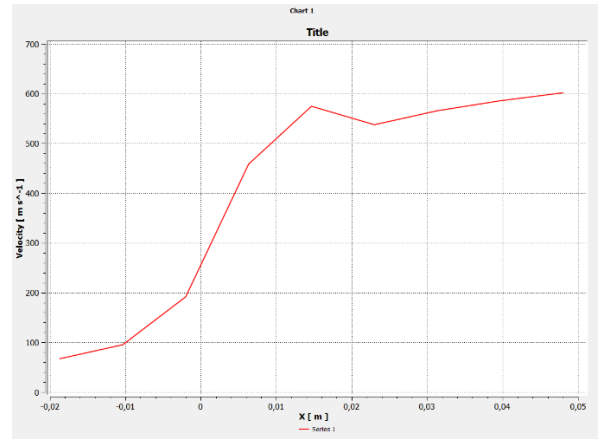


C-) Equal Inlet and Outlet Diameter

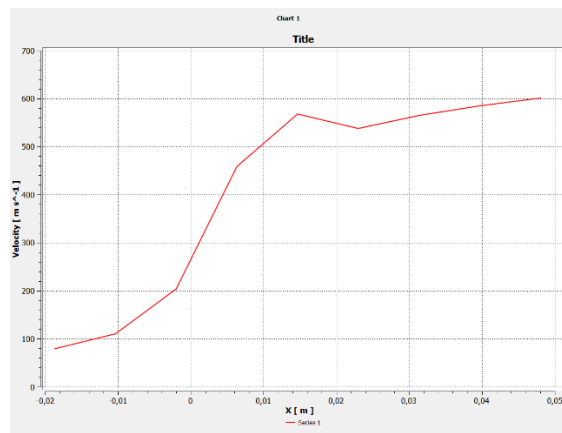
Figure 20: Velocity Vectors Different Inlet and Outlet Diameters



A-) Small Inlet Diameter



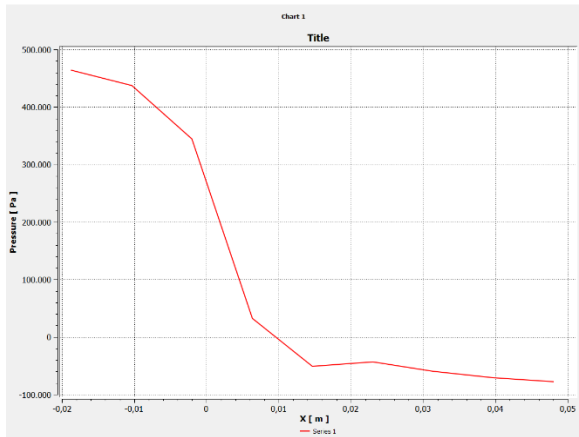
B-) Large Inlet Diameter Small Outlet Diameter



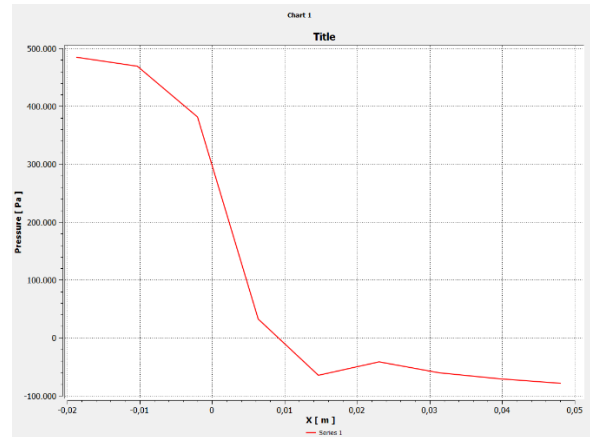
C-) Equal Inlet and Outlet Diameter

Figure 21: Velocity Charts Different Inlet and Outlet Diameters

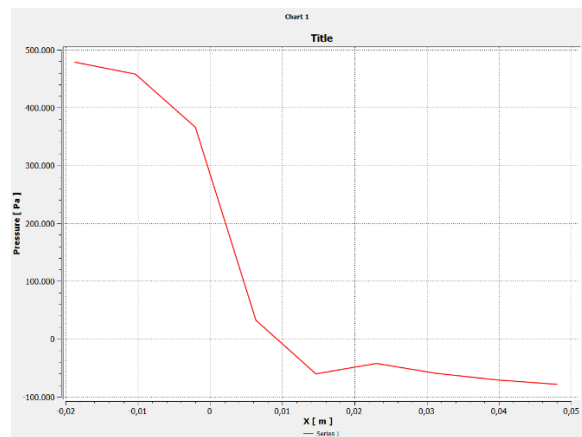
Comment: Due to the constant pressure at the entrances, according to the continuity equation ($A_1 V_1 = A_2 V_2$), the inlet velocity has changed inversely with the cross-sectional area.



A-) Small Inlet Diameter



B-) Large Inlet Diameter Small Outlet Diameter



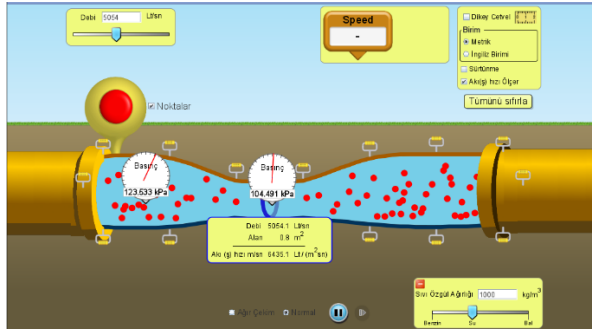
C-) Equal Inlet and Outlet Diameter

Figure 22: Pressure Charts Different Inlet and Outlet Diameters

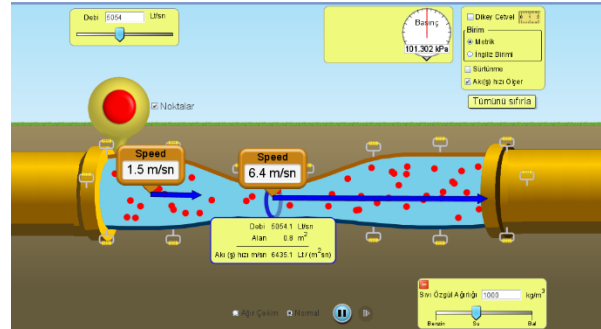
Comment: When the constant pressure at the entrances encountered a wider cross section, the pressure change started to occur earlier.

Simulations:

In addition to all the analyzes, we also examined the experiments in simulation. In the simulation, we re-examined the pressure and velocities of the fluid at the throat and inlet. We also observed the changes that occurred when we changed the type of fluid [4].



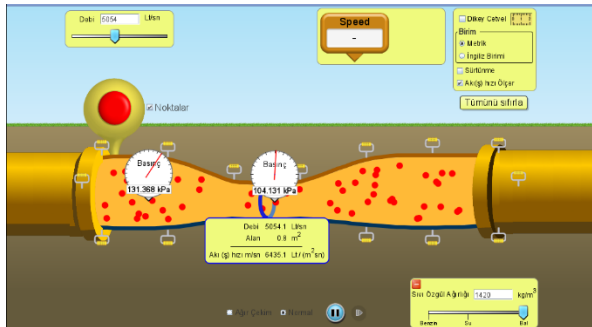
A-) Pressure



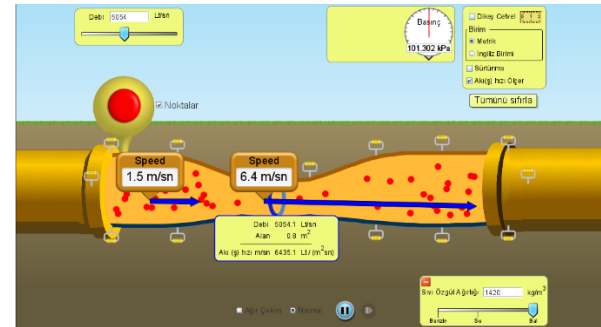
B-) Velocity

Figure 23: Venturi Meter Simulation (Flow type is water)

Comment: As can be seen, as the cross-sectional area through which the fluid passes decreases, the speed increases, and the pressure decreases.



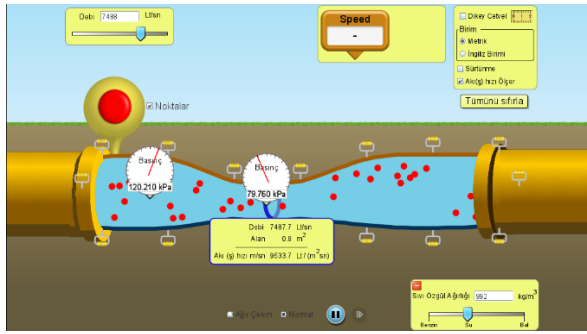
A-) Pressure



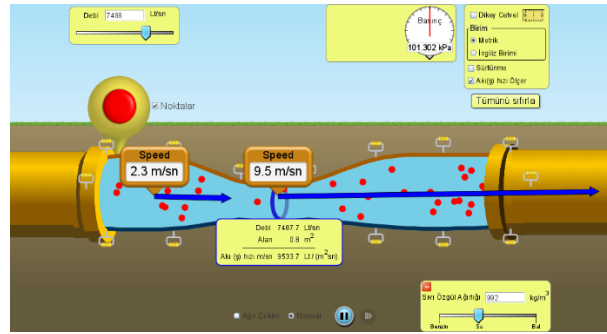
B-) Velocity

Figure 24: Venturi Meter Simulation (Flow type is honey)

Comment: When the density (specific gravity) of the fluid increases, the velocity does not change, while the pressure increases. We can observe this result in the ideal gas ($PV = nRT$) equation.



A-) Pressure



B-) Velocity

Figure 25: Difference Flow Rate in Venturi Meter Simulation (Flow type is water)

Comment: Here the pressure and velocity changes observed as the flow rate increases are simulated. As the flow rate increased, the speed increased in direct proportion according to the continuity equation. Pressure increased inversely with speed.

$$(Q = V_1 A_1 = V_2 A_2)$$

V.CONCLUSIONS

In this joint experimental study, when the fields of a venturi meter are kept constant, the experimental and real values measured by the change in the amount of flow per unit time are compared. In this comparison, the most accurate experimental experiment was found by using the discharge coefficient value. In addition, flow analysis of this venturi meter was performed in Ansys. At the same time, analysis of a different venturi meter in terms of geometry was performed in Ansys. In this way, the changes caused by the difference in geometry were examined. Finally, the analyzes of the venturi meter, which has the most accurate trial, and the venturi meter, which has different dimensions, are compared using the Bernoulli and Continuity equations. At the same time, these changes were observed in the simulation. Multiple trials have increased the accuracy.

According to the calculations and analysis made, the amount of increasing pressure head difference increases the amount of flow per unit time. Our flow accelerates and the pressure decreases in the throat area where the cross-sectional area is less. This result is proven in the Bernoulli equation. In addition to the results we have found, when we change the specific gravity of the fluid in the simulation, the velocity of the fluid remains constant, while the pressure of the fluid changes in direct proportion to the specific gravity. At the same time, the flow that we send laminar in simulation continues without disturbing its order when it comes across a different cross-sectional area.

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- [4] <https://phet.colorado.edu/sims/cheerpj/fluid-pressure-and-flow/latest/fluid-pressure-and-flow.html?simulation=fluid-pressure-and-flow&locale=tr>